5.2 Classwork: Average Velocity and Displacement — Solutions

1. A commuter train travels along a straight track from Station A to Station C, passing through Station B.

   - **Stage 1:** It travels East for 15 minutes at a constant speed of 80 km h^{-1}.
   - **Stage 2:** It stops at Station B for 10 minutes.
   - **Stage 3:** It continues East for another 12 km at a constant speed of 48 km h^{-1}.

   (a) Calculate the total displacement from Station A to Station C in kilometers. [2]

   (b) Determine the total time taken for the entire journey in hours. [2]

   (c) Calculate the **average velocity** for the entire journey in km h^{-1}. [2]

Solutions

(a) Stage 1 distance: $80 \times \frac{15}{60} = 80 \times 0.25 = 20$ km

Total displacement $= 20 + 12 = \mathbf{32}$ km (east)

(b) Stage 1: $\frac{15}{60} = \frac{1}{4}$ h Stage 2: $\frac{10}{60} = \frac{1}{6}$ h Stage 3: $\frac{12}{48} = \frac{1}{4}$ h

$$\text{Total} = \frac{1}{4} + \frac{1}{6} + \frac{1}{4} = \frac{3}{12} + \frac{2}{12} + \frac{3}{12} = \frac{8}{12} = \mathbf{\frac{2}{3}} \text{ h (40 minutes)}$$

(c) $$\text{Average velocity} = \frac{\text{displacement}}{\text{time}} = \frac{32}{2/3} = 32 \times \frac{3}{2} = \mathbf{48} \text{ km h}^{-1}$$

2. A drone is programmed to fly along a straight North-South line for a surveying mission.
   - It flies 800 m North in 40 seconds.
   - It hovers in place for 10 seconds to take a photo.
   - It then flies 1200 m South in 60 seconds.

   (a) Calculate the total **distance** traveled by the drone. [1]

   (b) Calculate the final **displacement** of the drone from its starting point. [2]

   (c) Determine the **average speed** of the drone for the total 110-second flight. [2]

   (d) Determine the **average velocity** of the drone for the total flight. [2]

Solutions

(a) Distance $= 800 + 1200 = \mathbf{2000}$ m

(b) Taking North as positive:
Displacement $= 800 - 1200 = -400$ m
$= \mathbf{400}$ m South

(c) Average speed $= \dfrac{\text{distance}}{\text{time}} = \dfrac{2000}{110} = \dfrac{200}{11} \approx \mathbf{18.2}\ \mathrm{m\,s^{-1}}$

(d) Average velocity $= \dfrac{\text{displacement}}{\text{time}} = \dfrac{-400}{110} = -\dfrac{40}{11} \approx \mathbf{-3.64}\ \mathrm{m\,s^{-1}}$
(i.e. $3.64\ \mathrm{m\,s^{-1}}$ South)

3. A professional cyclist completes a 20 km straight-line time trial.

   - For the first 10 km, she maintains an average velocity of v km h^{-1}.
   - For the final 10 km, she increases her effort and maintains an average velocity of $1.5v$ km h^{-1}.
   - The total time taken for the entire 20 km trial is 30 minutes.

   (a) Show that the time taken for the first half of the race is $\dfrac{10}{v}$ hours and the second half is $\dfrac{20}{3v}$ hours. [2]

   (b) Calculate the value of v. [3]

   (c) Calculate the cyclist's **average velocity** for the entire 20 km trip. [2]

   (d) Explain why the answer to part (c) is not simply the arithmetic mean of v and $1.5v$. [1]

Solutions

(a) First half: time $= \dfrac{\text{distance}}{\text{velocity}} = \dfrac{10}{v}$ hours

Second half: time $= \dfrac{10}{1.5v} = \dfrac{10}{\frac{3v}{2}} = \dfrac{10 \times 2}{3v} = \dfrac{20}{3v}$ hours

(b) Total time $= \dfrac{1}{2}$ hour (30 minutes)

$$\frac{10}{v} + \frac{20}{3v} = \frac{1}{2}$$

$$\frac{30}{3v} + \frac{20}{3v} = \frac{1}{2}$$

$$\frac{50}{3v} = \frac{1}{2} \Rightarrow 3v = 100 \Rightarrow v = \mathbf{\frac{100}{3}} \approx 33.3 \text{ km h}^{-1}$$

(c) Average velocity $= \dfrac{20}{1/2} = \mathbf{40}$ km h^{-1}

(d) The arithmetic mean of v and $1.5v$ is $1.25v \approx 41.7$ km h^{-1}. But the cyclist spent *more time* in the slower first half (equal distances, not equal times), so the average is weighted toward the slower speed.